CS 255 Business Requirements Document Template

System Components and Design

Purpose

What is the purpose of this project? Who is the client and what do they want their system to be able to do?

- The purpose of this project is to design a system for DriverPass that helps students prepare for driving tests.
- The system will allow users to take online practice exams and schedule in-person driving lessons.
- The client, DriverPass, wants a system that improves pass rates for driving tests and manages reservations, users, and training services.

System Background

What does DriverPass want the system to do? What is the problem they want to fix? What are the different components needed for this system?

- DriverPass identified that many students fail driving tests due to lack of proper preparation.
- The system will provide:
    - Online practice tests
    - Driving lesson scheduling
    - Customer and driver management
- The system must allow access from anywhere online and support different user roles.

Objectives and Goals

What should this system be able to do when it is completed? What measurable tasks need to be included in the system design to achieve this?

- Provide online practice exams and training materials
- Allow customers to schedule, cancel, and modify appointments
- Track reservations, drivers, vehicles, and customers
- Maintain up-to-date DMV rules and content
- Ensure secure access for different user roles (admin, customer, employee)
- Generate reports and track user activity

Requirements

Nonfunctional Requirements

In this section, you will detail the different nonfunctional requirements for the DriverPass system. You will need to think about the different things that the system needs to function properly.

Performance Requirements

What environments (web-based, application, etc.) does this system need to run in? How fast should the system run? How often should the system be updated?

- The system will be web-based and accessible on computers and mobile devices
- The system should load quickly and handle multiple users at the same time
- Updates from DMV should be reflected in real-time

Platform Constraints

What platforms (Windows, Unix, etc.) should the system run on? Does the back end require any tools, such as a database, to support this application?

- The system will run on web browsers (Chrome, Edge, etc.)
- It will be cloud-based to avoid local maintenance
- A database system is required to store users, appointments, and records

Accuracy and Precision

How will you distinguish between different users? Is the input case-sensitive? When should the system inform the admin of a problem?

- Each user will have a unique account and login
- The system will track all actions (create, update, delete)
- The system should notify admins of errors or suspicious activity

Adaptability

Can you make changes to the user (add/remove/modify) without changing code? How will the system adapt to platform updates? What type of access does the IT admin need?

- The system should allow adding or disabling training packages
- IT staff must be able to manage users and system settings
- The system should adapt to future updates and features

Security

What is required for the user to log in? How can you secure the connection or the data exchange between the client and the server? What should happen to the account if there is a "brute force" hacking attempt? What happens if the user forgets their password?

- Users must log in with username and password
- Password reset functionality must be available
- Different access levels (admin, employee, customer)
- Protect sensitive data (credit card info, personal data)
- Accounts may be locked after multiple failed login attempts

Functional Requirements

Using the information from the scenario, think about the different functions the system needs to provide. Each of your bullets should start with "The system shall . . ." For example, one functional requirement might be, "The system shall validate user credentials when logging in."

- The system shall allow users to create accounts
- The system shall allow users to log in and log out
- The system shall allow customers to schedule driving lessons

- The system shall allow users to cancel or modify appointments
- The system shall track which driver and car are assigned
- The system shall allow customers to take online practice tests
- The system shall display test results and progress
- The system shall allow admins to manage users and reset passwords
- The system shall allow employees to create reservations for customers
- The system shall generate reports of user activity
- The system shall update DMV-related content

User Interface

What are the needs of the interface? Who are the different users for this interface? What will each user need to be able to do through the interface? How will the user interact with the interface (mobile, browser, etc.)?

- The system will be web-based and accessible on mobile and desktop
- Users include:
    - Customers (schedule lessons, take tests)
    - Employees (manage appointments)
    - Admin/IT (full access and system control)
- Interface should include:
    - Forms for registration and scheduling
    - Dashboard showing test progress
    - Reservation calendar
    - Reports and activity logs

Assumptions

What things were not specifically addressed in your design above? What assumptions are you making in your design about the users or the technology they have?

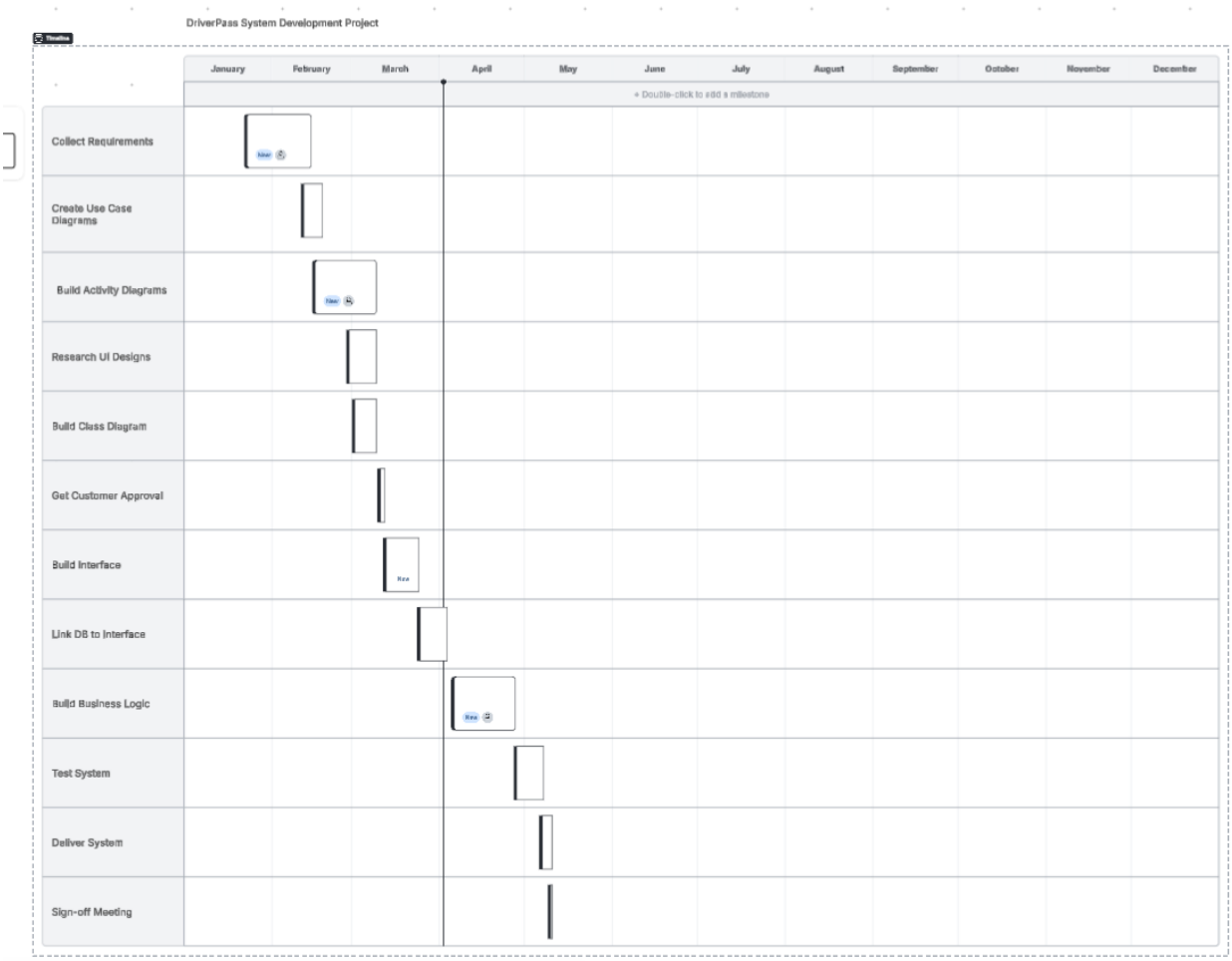
- Users have internet access
- Customers have basic knowledge of using web applications
- The DMV will provide updated information electronically
- Payment processing will be handled securely

Limitations

Any system you build will naturally have limitations. What limitations do you see in your system design? What limitations do you have as far as resources, time, budget, or technology?

- System depends on internet connection
- Changes to packages may require developer support

- Budget and time constraints may limit features
- Security risks always exist with online systems

Gantt Chart

Please include a screenshot of the GANTT chart that you created with Lucidchart. Be sure to check that it meets the plan described by the characters in the interview.